Enhanced Advancing Front Delaunay Meshing in TCAD

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Abstract **– New developments of the advancing front Delaunay mesher *deLink* are presented. The motivation for the advancing front Delaunay method and its differences compared to the in three dimensions commonly used incremental Delaunay algorithm are shortly discussed. A new element quality assessment with regard to anisotropy and so called "flat tets" or slivers is proposed.**

I. INTRODUCTION

The advancing front Delaunay method has been introduced in [1, 2]. This method is motivated by two important advantages. First, the advancing front Delaunay algorithm is a locally confined method similar to other advancing front methods. All other Delaunay methods are not, because they require the construction of a mesh for the structure surrounding convex hull (e.g. the bounding box). Secondly, it avoids expensive intersection tests which are typically employed for advancing front methods, by using a Delaunay kernel.

Figure 1 sketches these characteristics and depicts the difference between the commonly used incremental Delaunay algorithm [3] and the advancing front Delaunay method. The incremental method which is an example of a method which is locally not confined (Fig. 1-a) maintains a triangulation of the entire domain at all times. Points can be sequentially inserted in any order, and the boundary is recovered after all points have been inserted. Figure 1-b looks the same for advancing front methods with and without Delaunay kernels. However for methods with a Delaunay kernel all elements of the advancing front are Delaunay simplices and a collision detection is therefore reduced to checking for double elements (intersections in principle do not occur).

One immediate gain of the advancing front Delaunay method is the ability to confine the meshing to a local area should the need arise for mesh adaptation during transient simulation as for example in semiconductor diffusion simulation. Not only new mesh points can be introduced, but also point removal can be handled by local remeshing of cavities (Fig. 1-c and Fig. 1-d). The incremental Delaunay algorithm allows by nature point insertion, but not removal.

Alternatively coarsening can be performed by using hierarchical meshes where the solver stores all levels of refinement. However such hierarchical methods currently only apply simple refinement schemes and are not able to allow Steiner Point refinement [4].

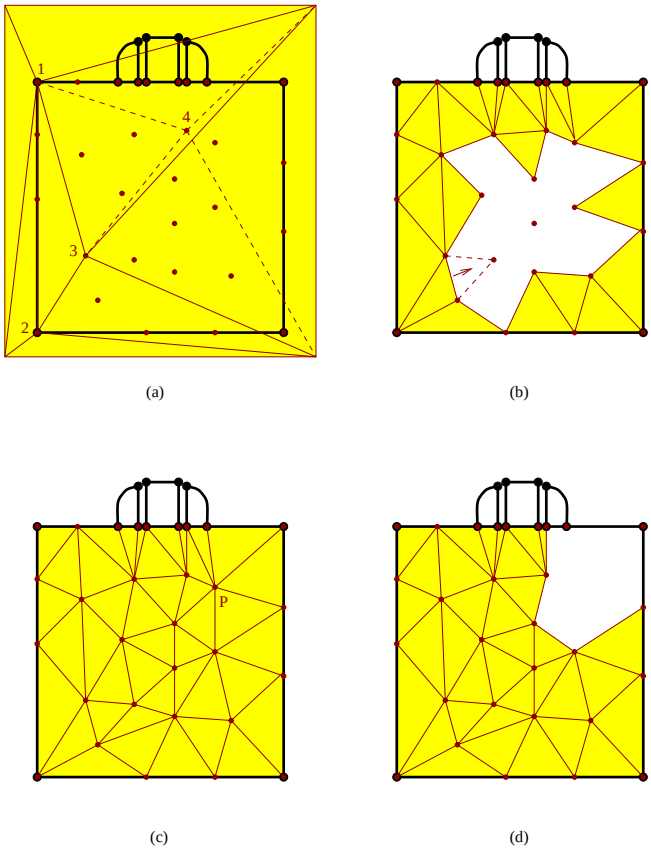

Figure 1: (a) Incremental Delaunay method with inserted points 1–3, point 4 will be inserted next. (b) Advancing front Delaunay method. (c) Point P is removed (coarsening). (d) Example of a non-convex cavity which evolves after point removal.

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Abstract **– New developments of the advancing front Delaunay mesher *deLink* are presented. The motivation for the advancing front Delaunay method and its differences compared to the in three dimensions commonly used incremental Delaunay algorithm are shortly discussed. A new element quality assessment with regard to anisotropy and so called "flat tets" or slivers is proposed.**

一、简介

前进前沿 Delaunay 方法已在[1, 2]中介绍。这种方法有两个重要的优点。首先，前进前沿Delaunay算法与其他前进前沿方法类似，是一种局部受限的方法。所有其他 Delaunay 方法都不是，因为它们需要为凸包周围的结构（例如边界框）构建网格。其次，它通过使用 Delauna y 内核避免了通常用于推进前端方法的昂贵的交叉测试。

图 1 概述了这些特征，并描述了常用的增量 Delaunay 算法 [3] 和先进的前沿 Delaunay 方法之间的差异。增量方法是局部不受约束的方法的一个例子（图1-a），它始终保持整个域的三角测量。点可以按任意顺序顺序插入，插入所有点后恢复边界。图 1-b 对于使用和不使用 Delaunay 核的先进前沿方法看起来是一样的。然而，对于具有 Delaunay 核的方法，前进前沿的所有元素都是 Delaunay 单纯形，因此碰撞检测被简化为检查双元素（原则上不会发生相交）。

先进的前沿 Delaunay 方法的一个直接好处是，如果在瞬态仿真期间（例如在半导体扩散仿真中）需要网格自适应，则能够将网格划分限制在局部区域。不仅可以引入新的网格点，还可以通过对空腔进行局部重新网格化来处理点移除（图1-c和图1-d）。这增量 Delaunay 算法本质上允许点插入，但不允许删除。

或者，可以通过使用分层网格来执行粗化，其中求解器存储所有级别的细化。然而，这种分层方法目前仅应用简单的细化方案，并且无法允许斯坦纳点细化[4]。

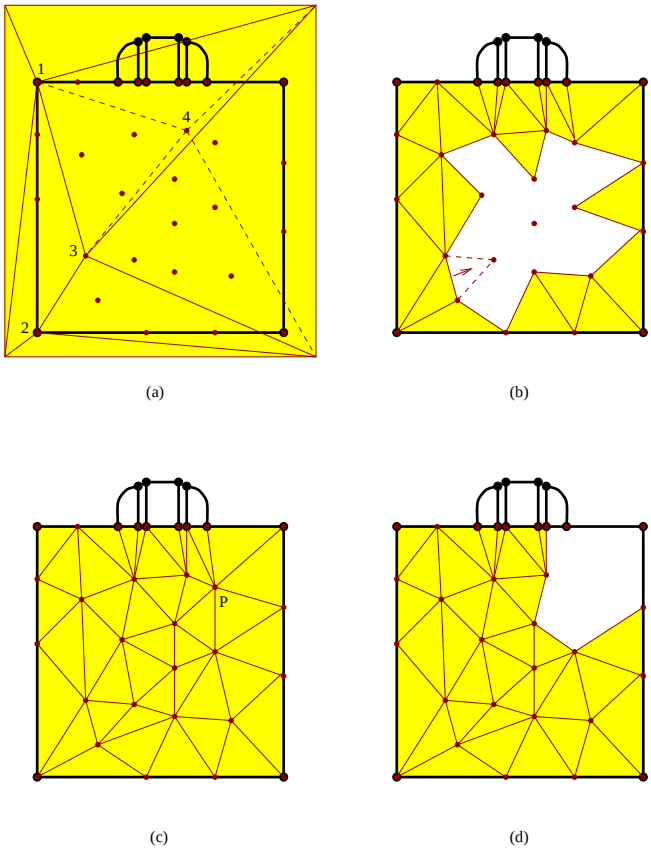

图 1：(a) 插入点 1{3 的增量 Delaunay 方法，接下来将插入点 4。 (b) 推进前沿Delaunay 方法。 (c) 点 P 被移除（粗化）。 (d) 点移除后形成的非凸空腔示例。

The discussed advantages of a combined advancing front and Delaunay method motivates its application to problems where Delaunay meshes are not required. While for finite volume discretization Delaunay meshes are required and sufficient, finite element methods motivate other quality enhancements [5]. While in two dimensions Delaunay meshes provide elements with good shape, in higher dimensions Delaunay meshes generally do not guarantee well shaped elements [6]. We present new improvements to the advancing front Delaunay method with regard to element shape quality, anisotropy, and slivers by means of Steiner point refinement.

II. PRESCRIBED ANISOTROPY

The applied state of the art in adaptive refinement methods with respect to e.g. a doping gradient is of isotropic nature. It remains an outstanding challenge to automatically adapt a mesh anisotropically and control the element quality at the same time. Edge refinement where each edge is considered separately may take the direction of the edge into account and hence should in principle be able to provide anisotropy. However, the resulting mesh quality is insufficient and a large number of obtuse angles is usually introduced. Alternatively, the refinement scheme can consider an entire element (triangle or tetrahedron) and upon refinement split the longest edge. In this way the element quality can be controlled but the refinement is isotropic again.

After ion implantation the doping is not necessarily smooth, nor has smooth gradients, and requires a highly anisotropic mesh density. A remedy to these issues is the less general approach to prescribe the desired anisotropy. It is of practical importance to construct an initial anisotropic mesh complying with mesh specifications provided by the application engineer. Such specifications include maximum element sizes given independently for different directions in a certain region.

We implemented the capability to provide such mesh spacing definitions, whereby non-coordinate-axis-aligned anisotropy and overlapping regions of spacing definitions are supported. From the possibly large set of spacing definitions for various regions one global background mesh will be extracted which holds the minimum density at each point complying with the specifications. In such a way it is fairly easy to pre-calculate a set of mesh points and thereby generate an initial anisotropic mesh. Furthermore, it is possible to achieve non-planar anisotropy near material interfaces. This is achieved by a special technique to compute the average surface normal and to allow the specification of a mesh spacing orthogonal to the material interface. However, at the moment the mesh connectivity of such boundary layers is not enforced. Such offsetting techniques have been exploited by [7].

The advancing front Delaunay method requires a priori surface meshing. The implemented surface module takes the pre-calculated points robustly into account. Points located outside the structure surface are ignored by the locally confined volume method. Points located *on* the surface are integrated into the surface triangulation resulting in a surface mesh. Points close to the surface can be snapped to the surface (projection) or deleted. The pre-calculation of points from the global background mesh avoids putting points too close to structure vertices, so no robustness issues evolve from this issue as well. Point location and snapping is performed using a point bucket octree.

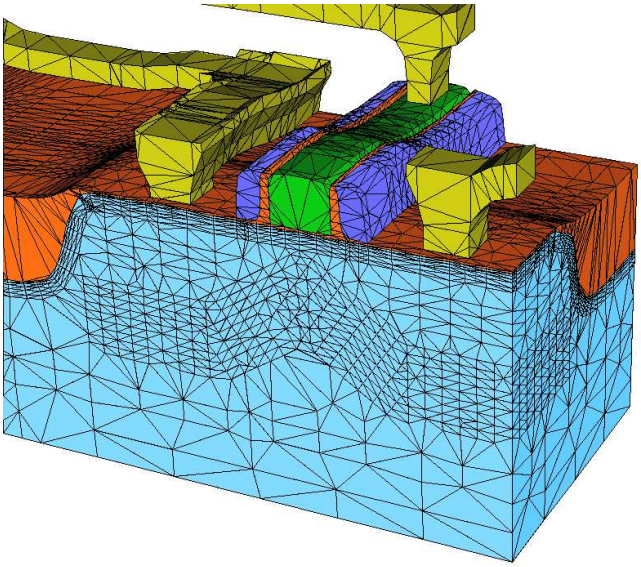

Figure 2: STI structure with 3D rounded corner and thin layer, anisotropic surface mesh (16888 triangles).

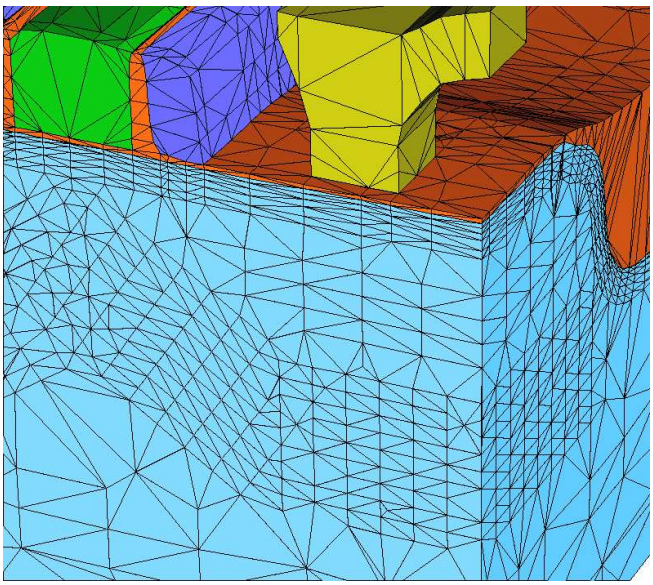

Figure 3: Region near silicon-oxide interface is shown enlarged.

所讨论的前进前沿和 Delaunay 方法相结合的优点促使其应用于不需要 Delaunay 网格的问题。虽然对于fi有限体积离散化来说，Delaunay 网格是必需的并且ffi足够了，但fi有限元方法激发了其他质量增强[5]。虽然在二维中，Delaunay 网格提供了具有良好形状的元素，但在更高维度中，Delaunay 网格通常不能保证形状良好的元素 [6]。我们通过 Steiner 点修正对先进的前沿 Delaunay 方法在单元形状质量、各向异性和条子方面提出了新的改进。

先进的前沿 Delaunay 方法需要先验的表面网格划分。实现的表面模块充分考虑了预先计算的点。局部约束体积法会忽略位于结构表面外部的点。位于表面 on 的点被集成到表面三角测量中，从而形成表面网格。靠近曲面的点可以捕捉到曲面（投影）或删除。从全局背景网格中预先计算点可以避免将点放置得太靠近结构顶点，因此这个问题也不会产生鲁棒性问题。点定位和捕捉是使用点桶八叉树执行的。

二.规定的各向异性

自适应细化方法中应用的最新技术掺杂梯度具有各向同性性质。自动各向异性地适应网格并同时控制单元质量仍然是一个突出的挑战。单独考虑每个边缘的边缘细化可以考虑边缘的方向，因此原则上应该能够提供各向异性。然而，所得的网格质量不足，并且通常引入大量钝角。或者，细化方案可以考虑整个元素（三角形或四面体），并在细化后分割最长的边。通过这种方式，可以控制单元质量，但细化又是各向同性的。

离子注入后，掺杂不一定是平滑的，也没有平滑的梯度，并且需要高度各向异性的网格密度。解决这些问题的方法是采用不太通用的方法来规定所需的各向异性。构建符合应用工程师提供的网格规范fi的初始各向异性网格具有实际意义。此类规范包括针对特定区域中的不同方向独立给出的最大单元尺寸。

我们实现了提供此类网格间距定义的功能，从而支持非坐标轴对齐的各向异性和间距定义的重叠区域。从各个区域的可能较大的一组间距定义中，将提取一个全局背景网格，该网格在符合规范的每个点处保持最小密度。通过这种方式，可以很容易地预先计算一组网格点，从而生成初始的各向异性网格。此外，可以在材料界面附近实现非平面各向异性。这是通过计算平均表面法线并允许指定与材料界面正交的网格间距的特殊技术来实现的。然而，目前此类边界层的网格连通性尚未强制执行。 [7]已经利用了这种偏移技术。

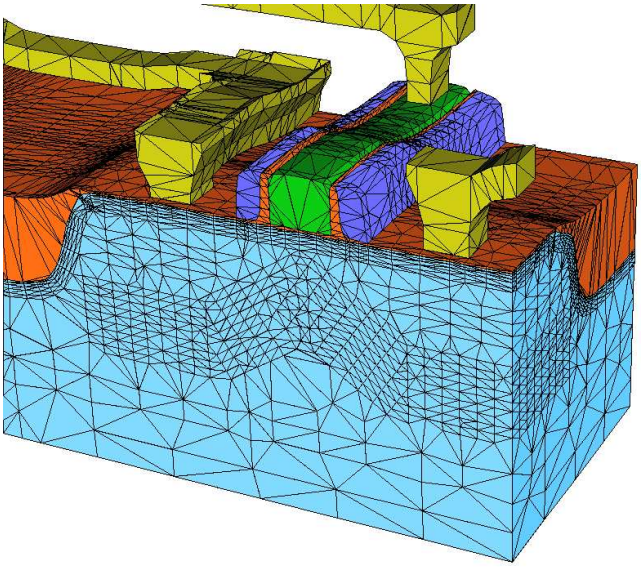

图 2：具有 3D 圆角和薄层、各向异性表面网格（16888 个三角形）的 STI 结构。

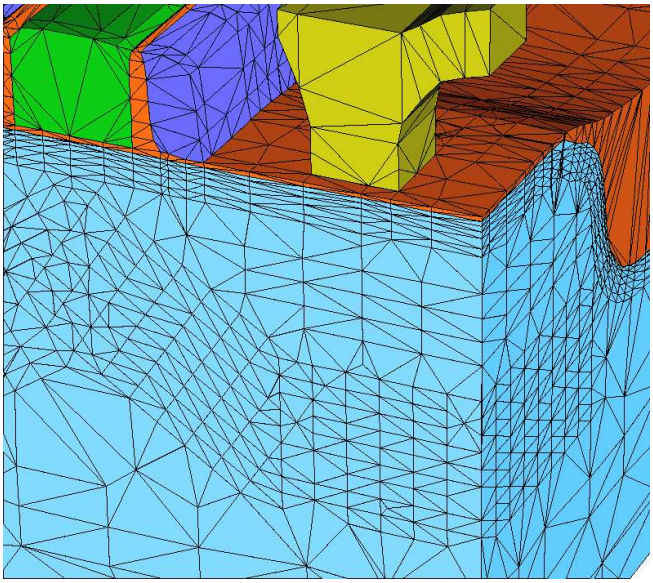

图 3：氧化硅界面附近的区域放大显示。

Figure 2 and Figure 3 show the surface mesh for an STI structure with a fully three-dimensional rounded corner. Anisotropy near the interface and in the bulk as well as the handling of thin layers (a form of geometry prescribed anisotropy, [8]) can be observed.

III. QUALITY ASSESSMENT

Slivers are three-dimensional elements with obtuse dihedral angle and which adversely affect, for example, finite element diffusion simulation [5]. Note that slivers do not exist in two dimensions. Other types of elements with limited shape quality are depicted in Fig. 4.

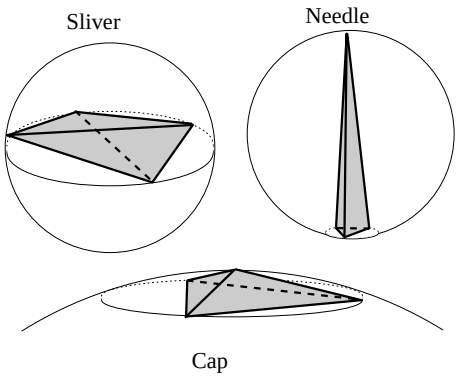

Figure 4: Fundamental element shapes in three dimensions.

Often, a single quality measure is employed to guide refinement methods to improve element quality. The measure itself is used to detect bad elements. Typical criteria for tetrahedra include

$$Q_s = \frac{l_{min}}{R} \tag{1}$$

$$AR = \frac{r}{R} \tag{2}$$

where R is the circumsphere radius, r the inner sphere radius, $lmin$ the minimum edge length, and AR the aspect ratio [9]. The problem with these criteria is that they cannot detect slivers (Q_s), or that they are not generally improved by e.g. Steiner point refinement (aspect ratio AR). Furthermore, most criteria are isotropic measures. They unnecessarily induce refinement for anisotropic elements such as needle elements which are actually desired.

We propose a new methodology whereby we first detect the *type* of element based on the number of acute and obtuse dihedral angles.

A *cap* is characterized by 3 acute and 3 obtuse dihedral angles. A *sliver* is characterized by 4 acute and 2 obtuse angles. A *needle* by our definition is characterized by 6 acute angles. Other combinations of angles indicate a mixture of the above shapes. For example, long, thin elements (needle-like) may contain 1 obtuse angle, 2 obtuse angles (sliver-like), or 3 obtuse angles (cap-like).

shape type	obtuse angles	acute angles	assessment	
			Q_s	AR
needle	0	6	low	low
cap	3	3	low	low
sliver	2	4	-	low

Table 1: Shape types, number of angles, and assessment.

The key idea is to apply different methods for different types, and to apply quality measures under the context of the type. Once the shape of an element is determined qualitatively, it is afterwards assessed quantitatively using a suitable measure. An example of type dependent quality assessment is a needle element. If a needle element has a high Q_s, AR, it is simply a normal, well shaped element. If it has a low Q_s, AR, it does not need refinement and indicates anisotropy. A cap with low Q_s can safely be refined using circumcenters as Steiner points. A sliver type element can be measured with AR and for low values requires healing.

IV. SLIVER REMOVAL

The order in which elements are processed is important, because e.g. Steiner point refinement to increase Q_s for a cap type element may produce new slivers. So sliver removal will be performed as a last step.

Currently, we perform healing of slivers with a mixed method resulting in a local transformation (tet-flip) or, if impossible, pair-wise collapsing of the 4 adjacent triangles onto each other with a following point insertion at the center of gravity to restore a consistent topology. This is still a heuristic approach and merits further investigation.

Figure 5 shows a visualization of slivers within an unoptimized volume mesh. Figure 6 and Figure 7 show the resulting volume mesh after sliver removal.

V. CONCLUSION

We implemented new features regarding anisotropy and sliver removal based on a new quality assessment methodology in *deLink*, the advancing front Delaunay mesher developed at TU Vienna. The resulting anisotropic and geometrically optimized volume mesh with improved element quality can be a basis for further doping refinement. Ideally, such a following refinement

图 2 和图 3 显示了具有全三维圆角的 STI 结构的表面网格。可以观察到界面附近和整体的各向异性以及薄层的处理（一种规定各向异性的几何形式，[8]）。

三．质量评估

条子是具有钝二面角的三维元素，它会对ff产生不利影响，例如fi-有限元ff扩散模拟[5]。请注意，条子不存在于二维空间中。其他类型的形状质量有限的单元如图 4 所示。

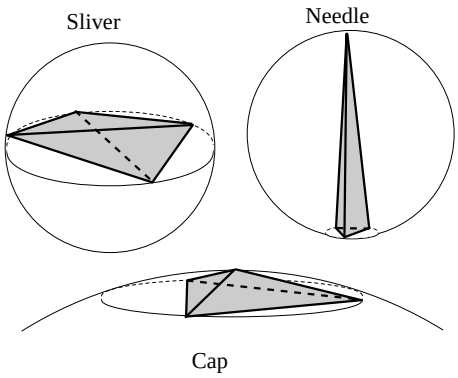

图 4：三个维度的基本元素形状。

通常，采用单一质量度量来指导改进方法以提高元素质量。该措施本身用于检测不良元素。四面体的典型标准包括

$$Q_s = \frac{l_{min}}{R} \tag{1}$$

$$AR = \frac{r}{R} \tag{2}$$

其中R是外接球半径，r是内球半径，$lmin$是最小边长，AR是长宽比[9]。这些标准的问题在于它们无法检测条子（Q_s），或者它们通常不会通过例如斯坦纳点修正fi（纵横比AR）。此外，大多数标准都是各向同性的措施。它们不必要地引起对各向异性元素（例如实际需要的针元素）的细化。

我们提出了一种新的方法，首先根据锐角和钝角的数量来检测元素的$type$。

cap 的特征是有 3 个锐角和 3 个钝角。 $sliver$ 的特征是 4 个锐音和 2 个

shape type	obtuse angles	acute angles	assessment Q_s	AR
needle	0	6	low	low
cap	3	3	low	low
sliver	2	4	-	low

表 1：形状类型、角度数量和评估。

钝角。根据我们的定义，$needle$ 的特征是有 6 个锐角。角度的其他组合表示上述形状的混合。例如，长而细的元件（针状）可以包含 1 个钝角、2 个钝角（条状）或 3 个钝角（帽状）。

关键思想是对不同类型应用不同的方法，并在该类型的上下文中应用质量度量。一旦定性地确定了元素的形状，随后就可以使用适当的措施对其进行定量评估。类型相关质量评估的一个例子是针元件。如果针元件具有较高的 Q_s, AR，那么它只是一个正常的、形状良好的元件。如果它的Q_s, AR较低，则不需要fi细化并表示各向异性。可以使用外心作为 Steiner 点来安全地修改 Q_s 较低的上限。条子类型元素可以使用 AR 进行测量，对于低值需要修复。

四．条子去除

处理元素的顺序很重要，因为例如斯坦纳点精修fi增加盖型元件的Q_s可能会产生新的条子。因此，去除棉条将作为最后一步进行。

目前，我们使用混合方法对条子进行修复，从而导致局部变换（tet-flip），或者如果不可能的话，将 4 个相邻三角形成对折叠到彼此上，并在重心处插入以下点以恢复一致的拓扑。这仍然是一种启发式方法，值得进一步研究。

图 5 显示了未优化体积网格内条子的可视化。图 6 和图 7 显示了去除条子后生成的体积网格。

五、结论

我们根据维也纳工业大学开发的先进的前 Delaunay 网格划分器 $deLink$ 中的新质量评估方法，实现了有关各向异性和条子去除的新功能。由此产生的各向异性和几何优化的体积网格以及改进的元素质量可以成为进一步掺杂细化的基础。理想情况下，这样的以下改进fi

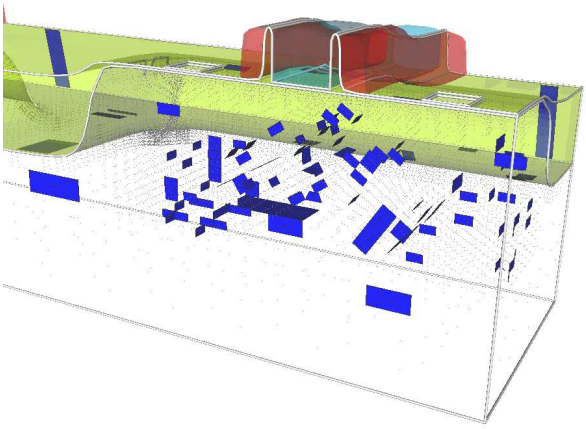

Figure 5: Visualization of slivers in an unoptimized volume mesh (131827 tets, 22585 points).

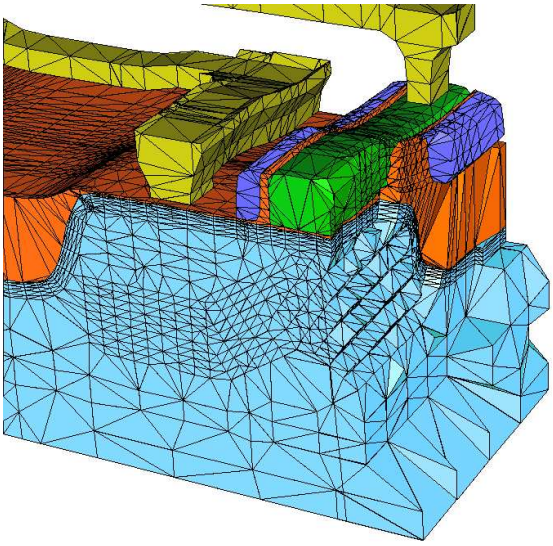

Figure 6: Optimized volume mesh visualized through clipping (131914 tets, 22622 points).

is limited to isotropic regions and does not destroy the prescribed anisotropy. Furthermore, such an adaptive doping refinement step should preserve the element quality especially if it is applied to finite element diffusion simulation.

ACKNOWLEDGEMENT

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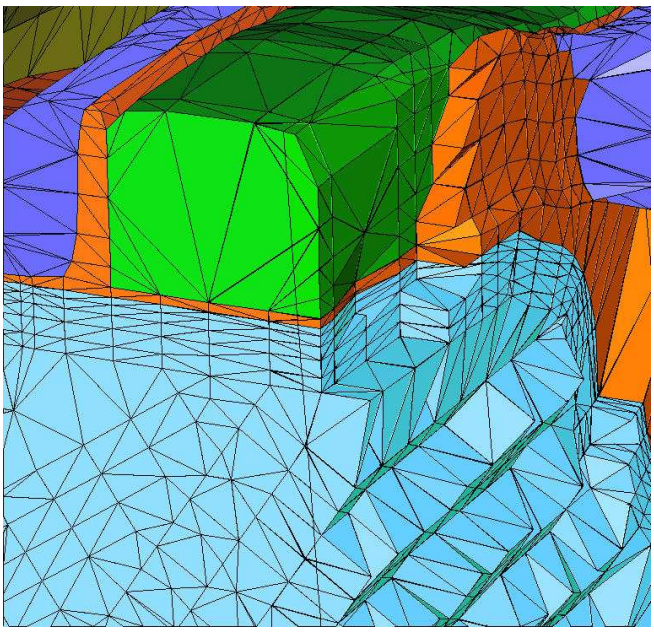

Figure 7: Optimized volume mesh enlarged.

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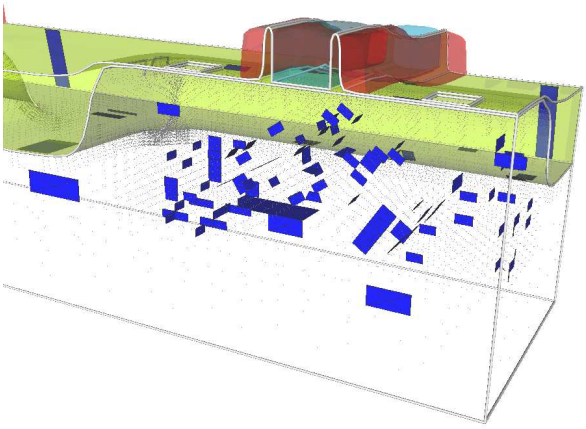

图 5：未优化体积网格中条子的可视化（131827 个四面体，22585 个点）。

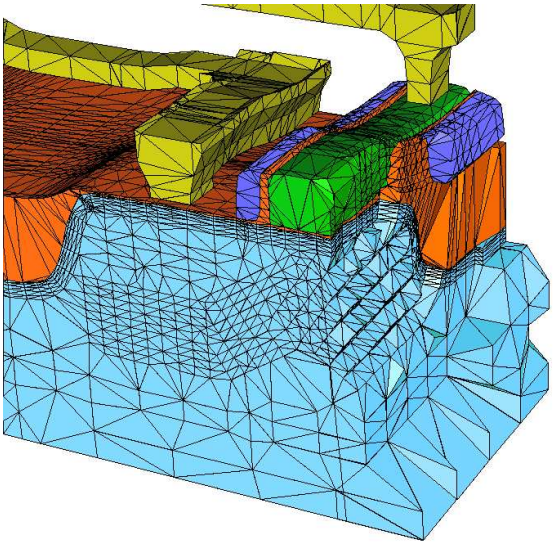

图 6：通过裁剪可视化的优化体积网格（131914 个四面体，22622 个点）。

仅限于各向同性区域并且不破坏规定的各向异性。此外，这种自适应掺杂精炼步骤应该保持元素质量，特别是当它应用于有限元素扩散模拟时。

致谢

我们承认这项工作的一部分是在 IST 项目 1999-11433 内进行的。

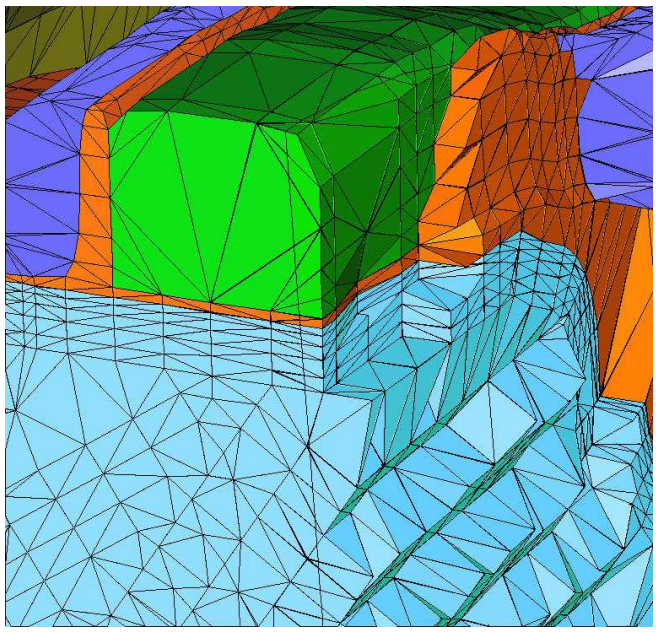

图 7：优化后的体积网格放大。

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